

Brief summary of the present ^{273}Ds and ^{269}Hs analysis

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Purpose

This note summarizes the current experimental evidence and tentative structural interpretation of two decay chains assigned to ^{273}Ds , observed in the $^{238}\text{U}(^{40}\text{Ar}, 5n)^{273}\text{Ds}$ reaction at the SHANS-2 separator. The manuscript is still under preparation, and the state assignments discussed below should be regarded as working hypotheses rather than final level assignments.

The main point on which we would appreciate expert advice is whether the lower-energy, longer-lived ^{273}Ds decay observed in the present work should be connected with the corresponding long-lived event reported in the DGFRS-2 experiment, or whether the two events should be kept separate at the present stage. This choice affects the estimated hindrance factor and the possible Nilsson-configuration assignment.

Experimental observations

The experiment was performed at SHANS-2 using the $^{238}\text{U}(^{40}\text{Ar}, xn)^{278-x}\text{Ds}$ reaction. At an average excitation energy of about 50 MeV, two complete decay chains assigned to the $5n$ channel were observed. The event-by-event decay sequences are shown in Fig. 1, and the corresponding decay properties are summarized together with previously reported chains in Table 1. The subsequent decays in both observed chains proceed through ^{269}Hs , ^{265}Sg , and terminal spontaneous fission of ^{261}Rf , and are consistent with previously reported decay properties of these nuclides[1].

The first chain starts with a higher-energy and shorter-lived ^{273}Ds alpha decay, whereas the second chain starts with a lower-energy and longer-lived decay. This two-branch pattern resembles that reported in the DGFRS-2 $^{40}\text{Ar} + ^{238}\text{U}$ experiment[2], but the present chains are complete and therefore provide additional information on the daughter decays.

Relation to the DGFRS-2 result

The long-lived ^{273}Ds alpha decay in the present work has

$$E_{\alpha} = 10.858(14) \text{ MeV}, \quad \Delta t = 8.320 \text{ ms.}$$

The corresponding long-lived event reported in the DGFRS-2 experiment [2] has

$$E_{\alpha} = 10.929(14) \text{ MeV}, \quad \Delta t = 41.703 \text{ ms.}$$

The energy difference is about 71 keV, which is larger than three times the present experimental energy uncertainty. Therefore, although both events belong to the lower-energy, longer-lived branch relative to the known high-energy ^{273}Ds decay, it is not obvious whether they should be assigned to the same state. The comparison of the available E_{α} and decay-time information in Fig. 2 illustrates this ambiguity, and we are considering both possibilities.

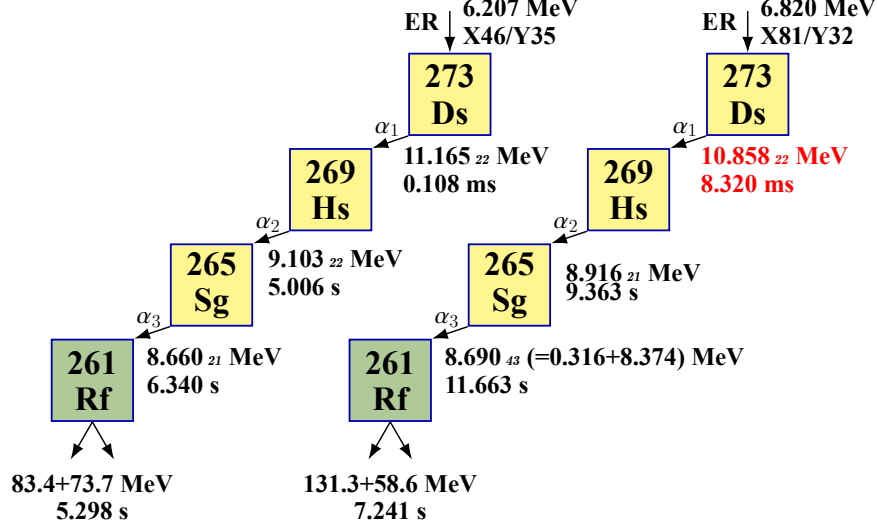


Figure 1: Two decay chains assigned to ^{273}Ds observed in the $^{238}\text{U}(^{40}\text{Ar}, 5n)^{273}\text{Ds}$ reaction. For each chain, the energy of the evaporation residue (ER) event and its X/Y position on the detector are given at the top, followed by the energies of the successive α -decay events and the terminal spontaneous-fission (SF) event, together with the time intervals between adjacent events. Energies in parentheses denote summed signals. Italic values indicate the corresponding energy uncertainties. Red text validated in the present work and the inferred assignments related to the isomeric states.

Possible low-lying state in ^{269}Hs

The daughter decay in the long-lived ^{273}Ds chain has $E_\alpha = 8.916 \text{ MeV}$, which is lower than the higher-energy ^{269}Hs alpha group near 9.0–9.3 MeV. In the present analysis, the lower-energy ^{269}Hs group is preferentially associated with the lower-energy, longer-lived ^{273}Ds branch, whereas the higher-energy ^{269}Hs group is mainly associated with the higher-energy, shorter-lived ^{273}Ds branch. No clear cross feeding has been observed within the available statistics.

Table 1: Decay properties of the observed chains assigned to ^{277}Cn , ^{273}Ds , ^{269}Hs , ^{265}Sg , and ^{261}Rf . The decay times of individual events are listed.

No.	Reference	^{277}Cn		^{273}Ds		^{269}Hs		^{265}Sg		^{261}Rf	
		E_α	Δt (ms)	E_α	Δt (ms)	E_α	Δt (s)	E_α	Δt (s)	E_α	Δt (s)
Ch1	[3]	11.45	0.280	11.08	0.110	9.23	19.7	(4.60)	7.4	8.52	4.7
Ch2	[3]	11.17	1.406	11.20	0.310	9.18	22.0	(0.2)	18.8	(SF)	14.5
Ch3	[4]	11.09	1.10	11.14	0.520	9.17	14.2	8.71	23.0	(SF)	2.97
Ch4	[4]	11.32	1.22	11.15	0.0399	9.25	0.270	8.70	79.0	(SF)	8.30
Ch5	[5]	11.07	0.370	11.03	0.373	9.15	36.0	8.66	13.8	(SF)	3.73
Ch6	[2]	–	–	11.017	0.184	–	–	8.397	9.0766	(SF)	3.0344
Ch7	[2]	–	–	10.929	41.703	(0.342)	4.0978	8.509	58.19	(SF)	0.3923
Ch8	This work	–	–	11.165	0.108	9.103	5.006	8.660	6.340	(SF)	5.298
Ch9	This work	–	–	10.858	8.320	8.916	9.363	8.690	11.663	(SF)	7.241

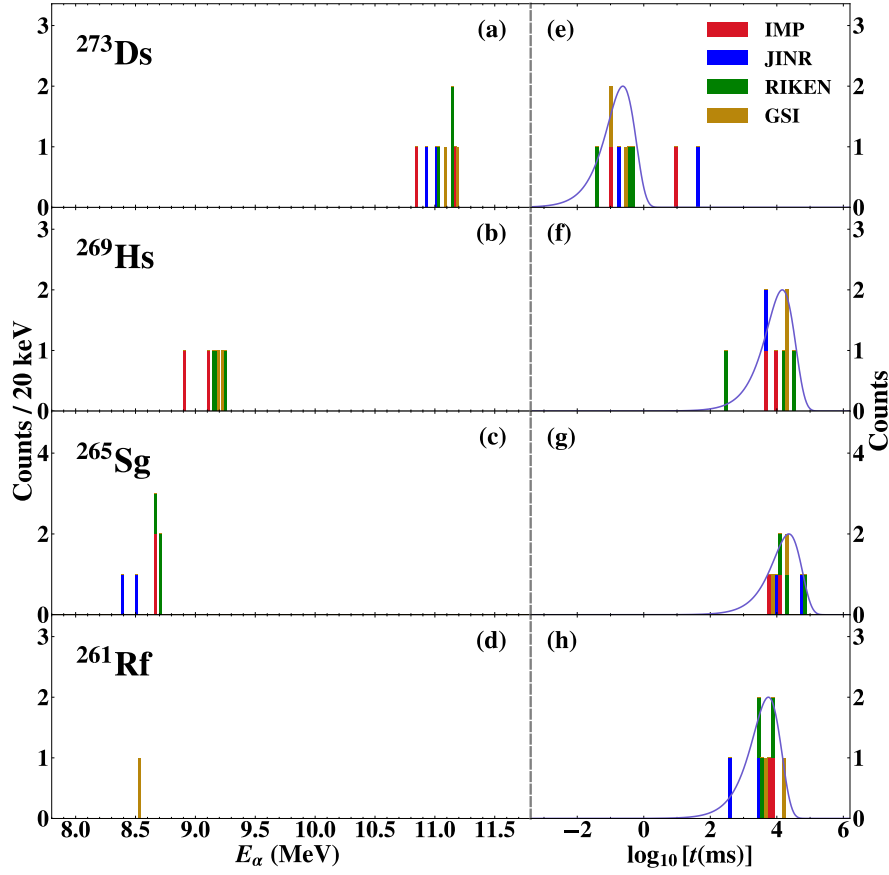


Figure 2: Comparison of the E_α and half-life information for decay chains involving ^{273}Ds . The red, blue, green, and yellow symbols correspond to the IMP, JINR[2], RIKEN[5, 4], and GSI[3, 6]. The fit curves in the decay-time panels are drawn according to Ref. [7].

This apparent selectivity in the α -decay feeding makes it more plausible that the 8.92–9.00 MeV group corresponds to an additional low-lying or isomeric state in ^{269}Hs , rather than merely to an unresolved α -decay branch from the same state. The compiled ^{269}Hs α -energy distribution in Fig. 3 shows how the present low-energy event compares with the directly produced and daughter-populated ^{269}Hs data. Owing to the inability of previous radiochemical experiments to determine the ER– α_1 correlation time, the present work provides the first direct lifetime measurement for this low-energy α -decay group in ^{269}Hs . However, its measured lifetime does not appear to differ significantly from those of the other groups. Given the still limited number of events, the proposed classification of states in ^{269}Hs should therefore remain tentative.

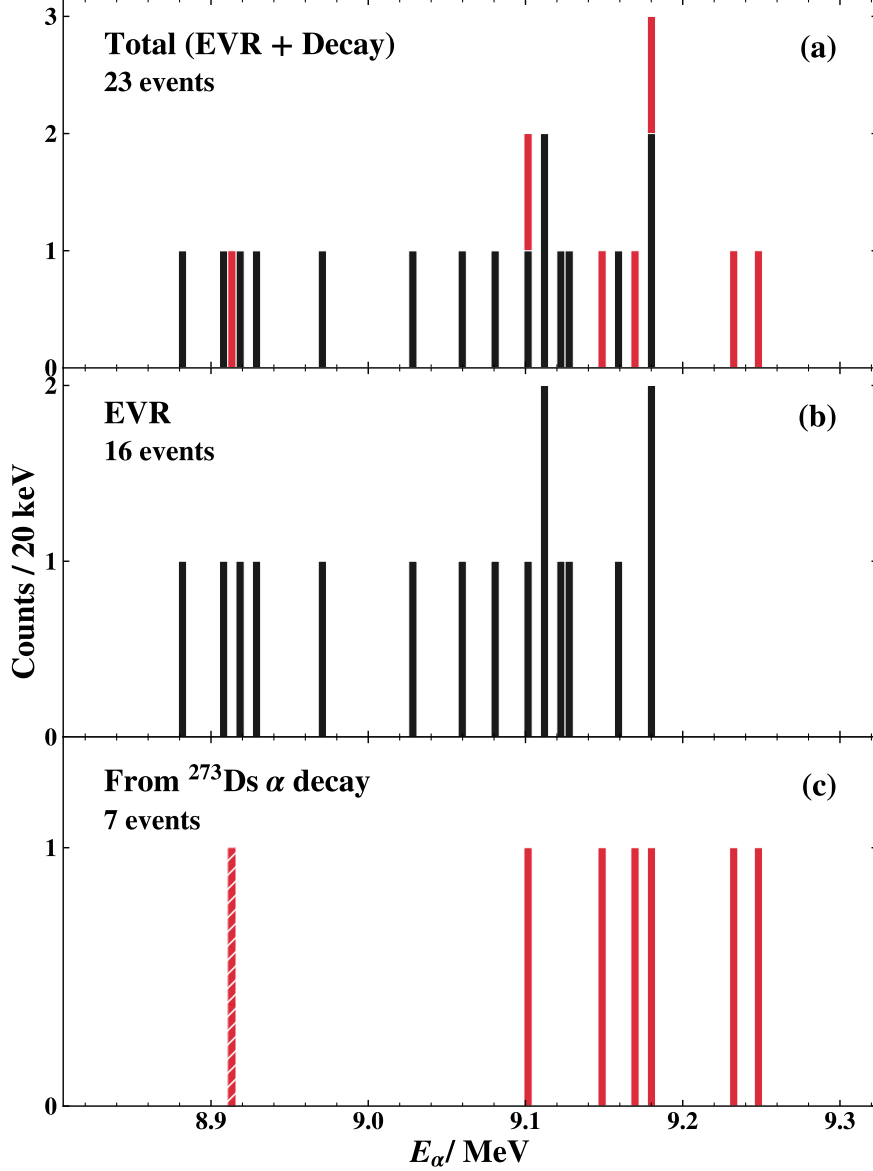


Figure 3: Energy distributions of α particles assigned to ^{269}Hs . The upper panel shows all events, the middle panel shows nuclei produced directly as evaporation residues (EVRs)[8, 9, 10, 11], and the lower panel shows nuclei populated through the α decay of ^{273}Ds [3, 4, 5]. The hatched bin indicates the event observed in the present work at $E_\alpha = 8.915$ MeV.

Tentative structural interpretation

The interpretation is guided by configuration-constrained potential-energy-surface calculations in the deformation space including β_2 , β_4 , and β_6 , followed by a macroscopic-microscopic mean-field analysis of relevant one-quasineutron configurations in ^{273}Ds and ^{269}Hs . The hindrance factors used to separate the long- and short-lived ^{273}Ds components are listed in Table 2, and the calculated one-quasineutron configurations entering the comparison are summarized in Table 3. The aim is to compare the observed alpha-decay branches with low-lying Nilsson configurations around the deformed $N = 162$ shell closure.

In this region, ^{273}Ds can be viewed as containing one neutron above the $N = 162$ gap, while

Table 2: Decay properties assigned to the long- and short-lived states of ^{273}Ds . The half-life of $^{273}\text{Ds}^a$ was obtained from a combined analysis of the long-lived decay events observed at JINR and IMP, whereas $^{273}\text{Ds}^b$ denotes the short-lived component. The value for $^{273}\text{Ds}^c$ was deduced from the single long-lived event observed at IMP only.

Nucleus	$T_{1/2}$	E_α^a (MeV)	HF [12, 13]	HF [14]
$^{273}\text{Ds}^a$	$17.34^{+31.55}_{-6.83}$ ms	10.89(2)	97^{+183}_{-40}	101^{+192}_{-42}
$^{273}\text{Ds}^b$	$0.16^{+0.08}_{-0.05}$ ms	11.17(2)	$3.8^{+1.9}_{-1.2}$	$4.2^{+2.1}_{-1.3}$
$^{273}\text{Ds}^c$	$5.76^{+27.62}_{-2.64}$ ms	10.86(2)	27^{+130}_{-13}	28^{+135}_{-13}

Table 3: Calculated one-quasineutron configurations and excitation energies for Ds and Hs.

Ds $(\beta_2, \beta_4, \beta_6) = (0.215, -0.076, -0.006)$		Hs $(\beta_2, \beta_4, \beta_6) = (0.231, -0.062, -0.012)$	
Configuration	E_x (MeV)	Configuration	E_x (MeV)
$3d_{5/2} \uparrow$: $3/2^+$ [622]	0.7387	$1i_{11/2} \downarrow$: $7/2^+$ [613]	0.0796
$1i_{11/2} \downarrow$: $9/2^+$ [604]	0.6655	$1j_{15/2} \downarrow$: $11/2^-$ [725]	0.0397
$3d_{5/2} \uparrow$: $5/2^+$ [613]	0.4730	$2g_{7/2} \downarrow$: $1/2^+$ [620]	0.0292
$2g_{7/2} \downarrow$: $3/2^+$ [611]	0.3192	$2g_{9/2} \uparrow$: $9/2^+$ [615]	0.0058
$1j_{15/2} \downarrow$: $13/2^-$ [716]	0	$3d_{5/2} \uparrow$: $3/2^+$ [622]	0

^{269}Hs contains one neutron hole below it. The long-lived ^{273}Ds branch may therefore reflect a high- Ω one-quasineutron configuration whose alpha decay is hindered by configuration mismatch and, possibly, by angular-momentum and parity changes.

Two working scenarios are being considered, as illustrated in Fig. 4.

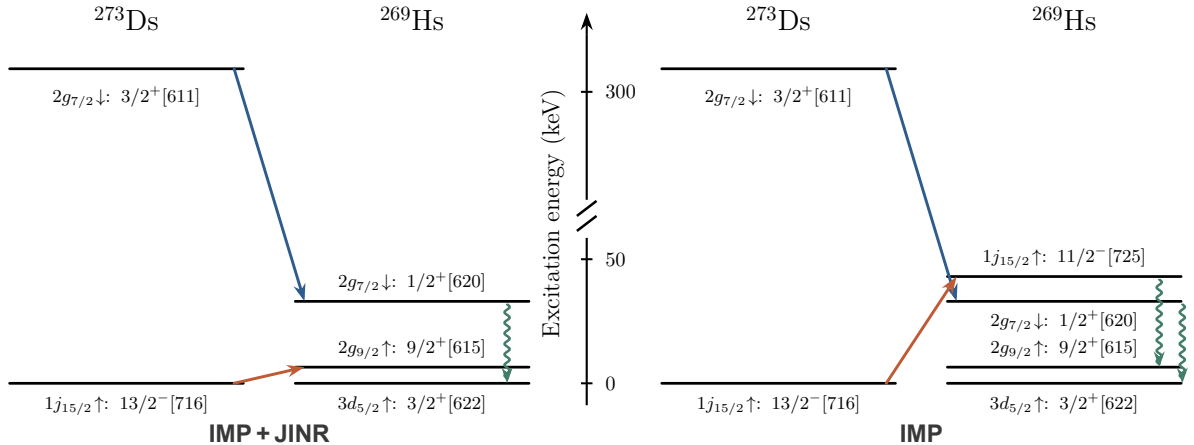


Figure 4: Comparison of the level assignments for the observed α -decay events. In the IMP+JINR interpretation (left), the two events are assigned to a common state in ^{269}Hs . In the IMP interpretation (right), the IMP event is assigned to a separate state in ^{269}Hs , followed by the indicated γ transition to the lower-lying state. The blue arrows denote the short-lived decay transitions, whereas the red arrows denote the long-lived decay transitions.

At present, neither the hindrance-factor analysis nor the configuration-constrained calculation

provides a unique assignment. They are used mainly as consistency checks for possible structural interpretations.

Systematics and limitations

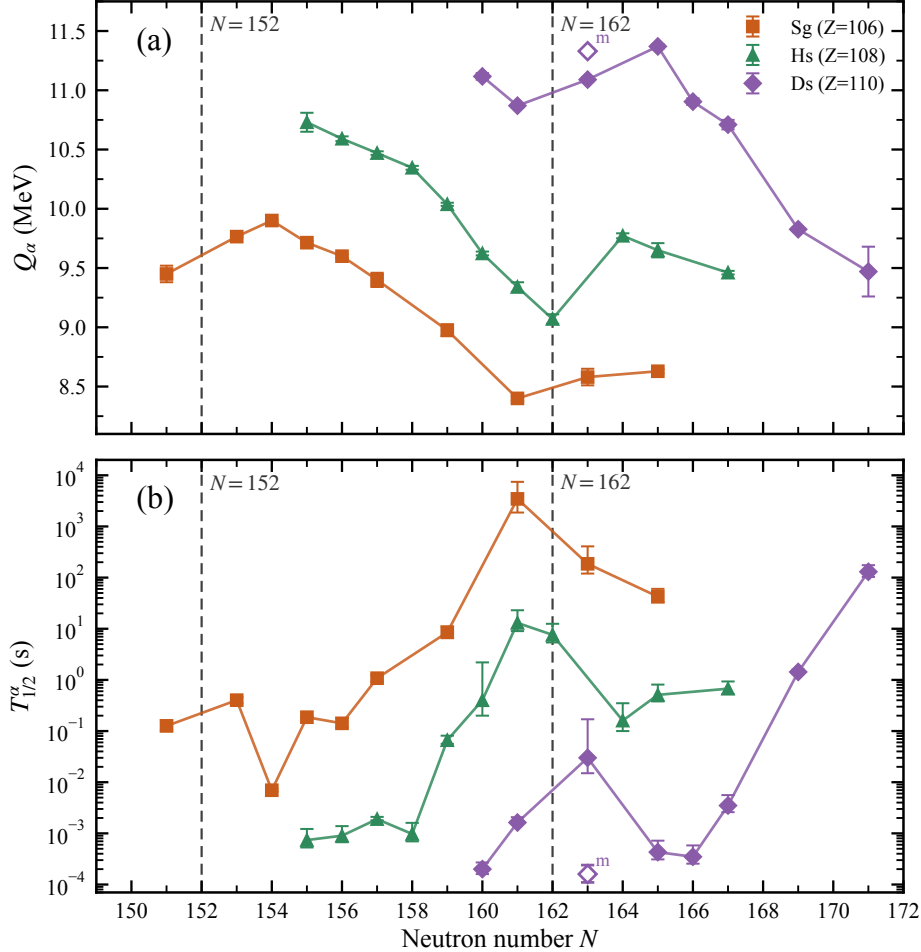


Figure 5: Systematics of Q_α values and partial α -decay half-lives for the Sg, Hs, and Ds isotopic chains. The dashed vertical lines mark $N = 152$ and $N = 162$. The open diamond labeled m represents $^{273}\text{Ds}^a$, whereas the filled diamond connected to the Ds isotopic trend represents $^{273}\text{Ds}^c$.

The isotopic systematics provide a useful broader context for the tentative assignments discussed above. As shown in Fig. 5, the Q_α values and partial α -decay half-lives in the Sg, Hs, and Ds chains do not evolve as a simple monotonic sequence across the deformed $N = 162$ region. The local behavior of the Hs and Ds chains is therefore qualitatively compatible with the possibility that different low-lying configurations contribute to the observed decay branches. In particular, the separation between the lower-energy, longer-lived ^{273}Ds branch and the higher-energy, shorter-lived branch is not inconsistent with a configuration-dependent α decay picture near the deformed shell closure.

This argument should remain modest. The same region is experimentally constrained by very small fusion-evaporation cross sections, so several key nuclei are represented by only one or a few correlated decay chains, and some historical chains are incomplete. As a result, adopted half-

lives, partial α -decay half-lives, and even branch assignments may be affected by low statistics, unresolved feeding, and assumptions about α branching ratios. The systematics in Fig. 5 should therefore be treated as a consistency check that supports the plausibility of the proposed picture, not as an independent proof of a particular level ordering or Nilsson-configuration assignment. More complete, higher-statistics decay data are needed to fix the ordering in ^{273}Ds and ^{269}Hs .

Open points for discussion

We would be grateful for comments on the following points.

1. From an experimental point of view, should the 10.858 MeV and 10.929 MeV long-lived ^{273}Ds events be assigned to the same state, or is it better to keep them separate until more data are available?
2. Can the selective association between the 8.92–9.00 MeV ^{269}Hs decay and the long-lived ^{273}Ds branch be regarded as meaningful evidence for an additional low-lying or isomeric state in ^{269}Hs ?
3. How reliable are hindrance factors and configuration-constrained calculations for assigning Nilsson configurations in odd- A superheavy nuclei with such limited statistics?
4. Are the following configuration assignments for the α -decay transitions physically reasonable,

$$\nu 13/2^- [716] \xrightarrow{\alpha} \nu 9/2^+ [615], \quad \text{HF} \sim 100$$

$$\nu 13/2^- [716] \xrightarrow{\alpha} \nu 11/2^- [725] \xrightarrow{\gamma} \nu 9/2^+ [615], \quad \text{HF} \sim 30$$

and

$$\nu 3/2^+ [611] \xrightarrow{\alpha} \nu 1/2^+ [620] \xrightarrow{\gamma} \nu 3/2^+ [622], \quad \text{HF} \sim 4$$

or should alternative configurations be considered?

5. The present calculations tend to favor assigning the lower-energy, longer-lived ^{273}Ds sequence to the ground state and the higher-energy, shorter-lived sequence to a low-lying excited state. Because the available systematics of Q_α values and half-lives near $N \approx 162$ are sparse, would such an ordering be reasonable, or would a more cautious interpretation without a definite ground-state assignment be preferable?

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